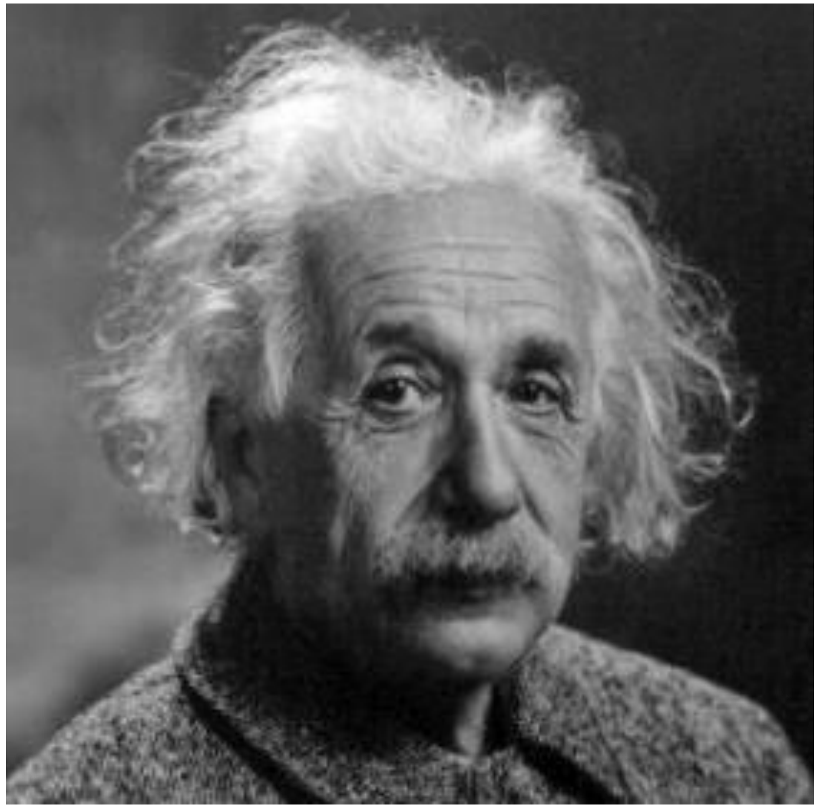
Ask the Right Questions to Explore Wisely

CSC105

Prof. Earl

Albert Einstein (1879-1955)



"If I had an hour to solve a problem and my life depended on it, I would use the first 55 minutes determining the proper questions to ask"

John Tukey (1915-2000)

“Far better an approximate answer to the right question, which is often vague, than the exact answer to the wrong question, which can always be made precise.”

Stephen Few – Aptitudes and Attitudes

- **Interested**
  - Having a genuine interest in the data. Interest can be increased, but not artificially. Pick topics you have an interest in.
- **Curious**
  - Maybe not interested in the exact data, but might want to see where the data takes you.
- **Self-Motivated**
  - A "get-up and go" attitude. Not waiting to be told what to do and go looking for questions to answer
- **Open-Minded**
  - Willingness to accept what you find
  - Willingness to discover that you are wrong

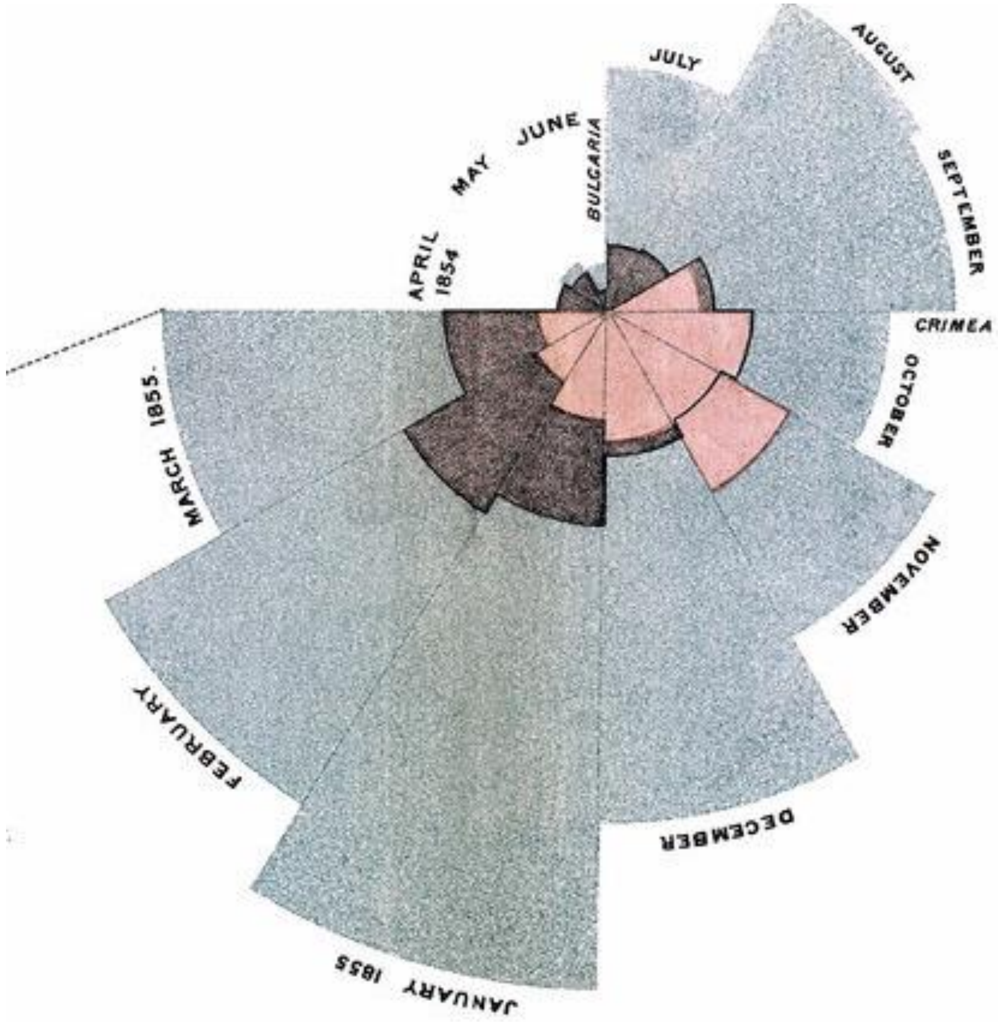
Stephen Few – Aptitudes and Attitudes

- **Awareness of What's Worthwhile**
  - What will provide a good return on investment. Having questions is good, but don't overload yourself with things that will take time with little return.
- **Methodical**
  - It's best to learn the best steps and then repeat them, rather than "reinventing the wheel" every time.
- **Creative**
  - "What if I try this?"
- **Skeptical**
  - Pay attention to the voice in our head that says something might be off. Try taking a step back and re-exploring.

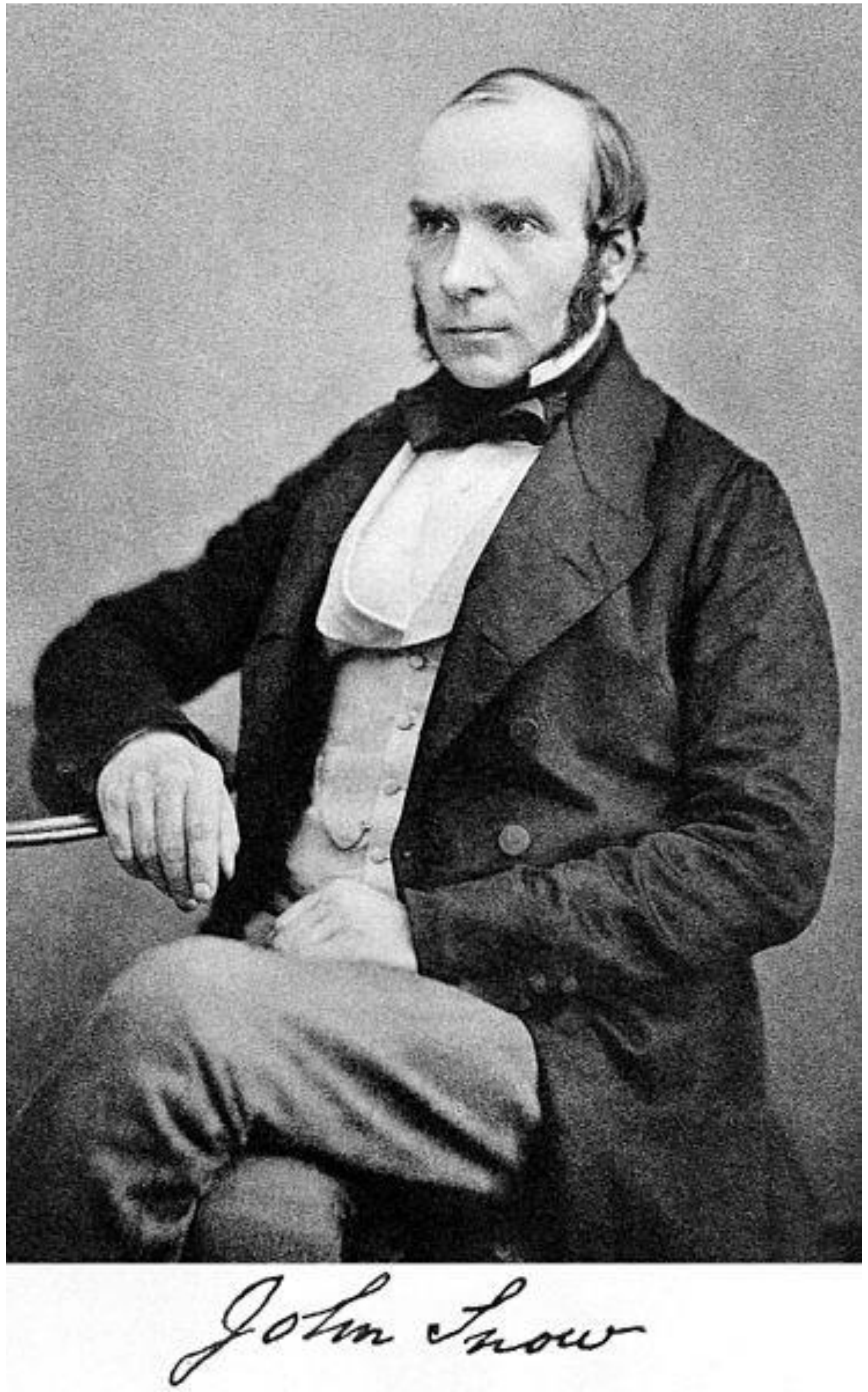
Stephen Few – Aptitudes and Attitudes

- **Honest**
  - Being truthful with your data, even when outside pressure wants a certain result.
- **Attentive**
  - Finding a quiet and distraction free to think about the data.
- **Analytical**
  - The ability to break a large complex problem into it's component parts.
- **Synthetical**
  - Not losing track of the big picture while in the component parts.
- **Good Communications**
- **Domain Knowledge**
- **Multidisciplinary Thinking**

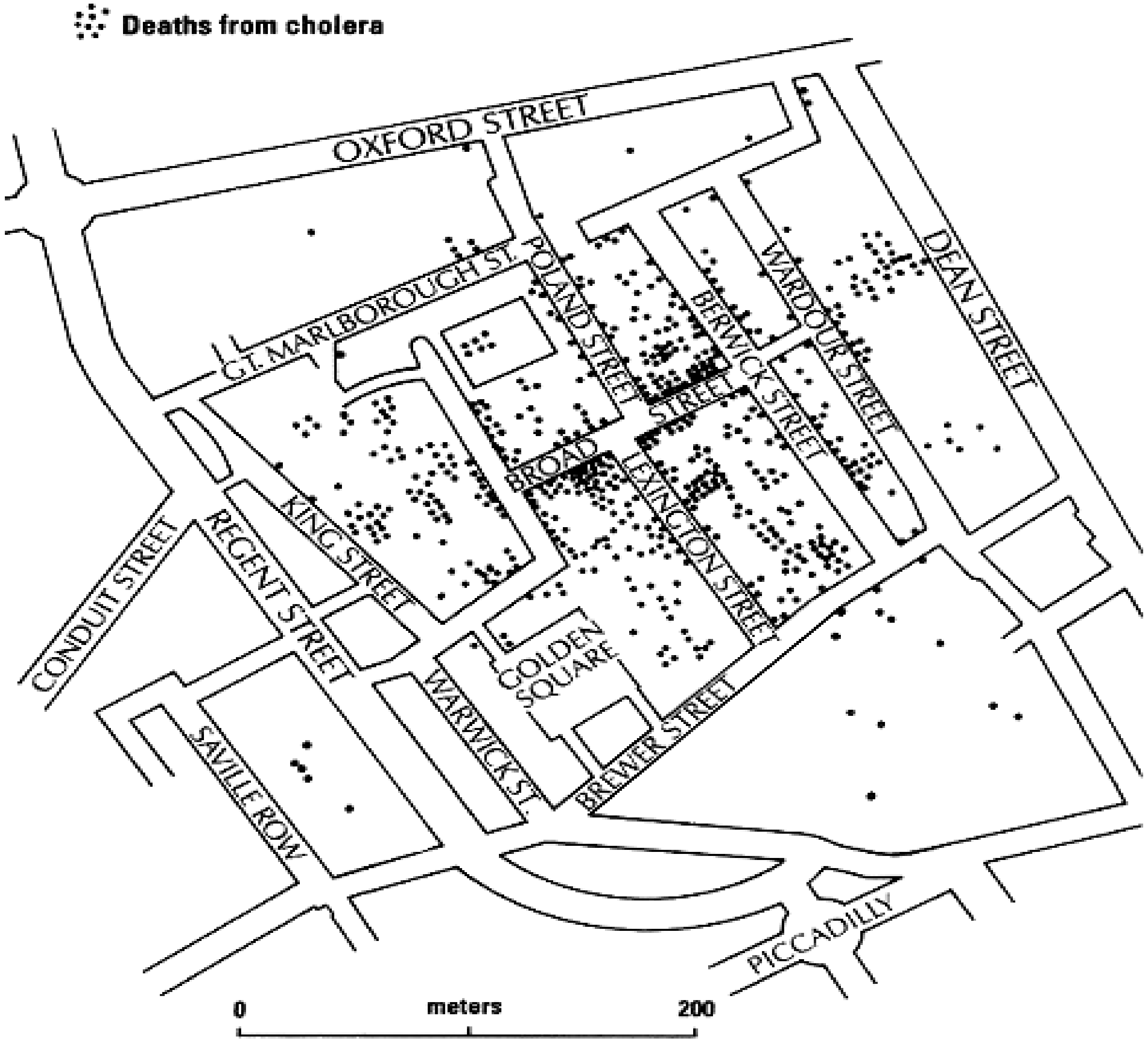
Florence Nightingale (1820-1910)
She asked great questions!



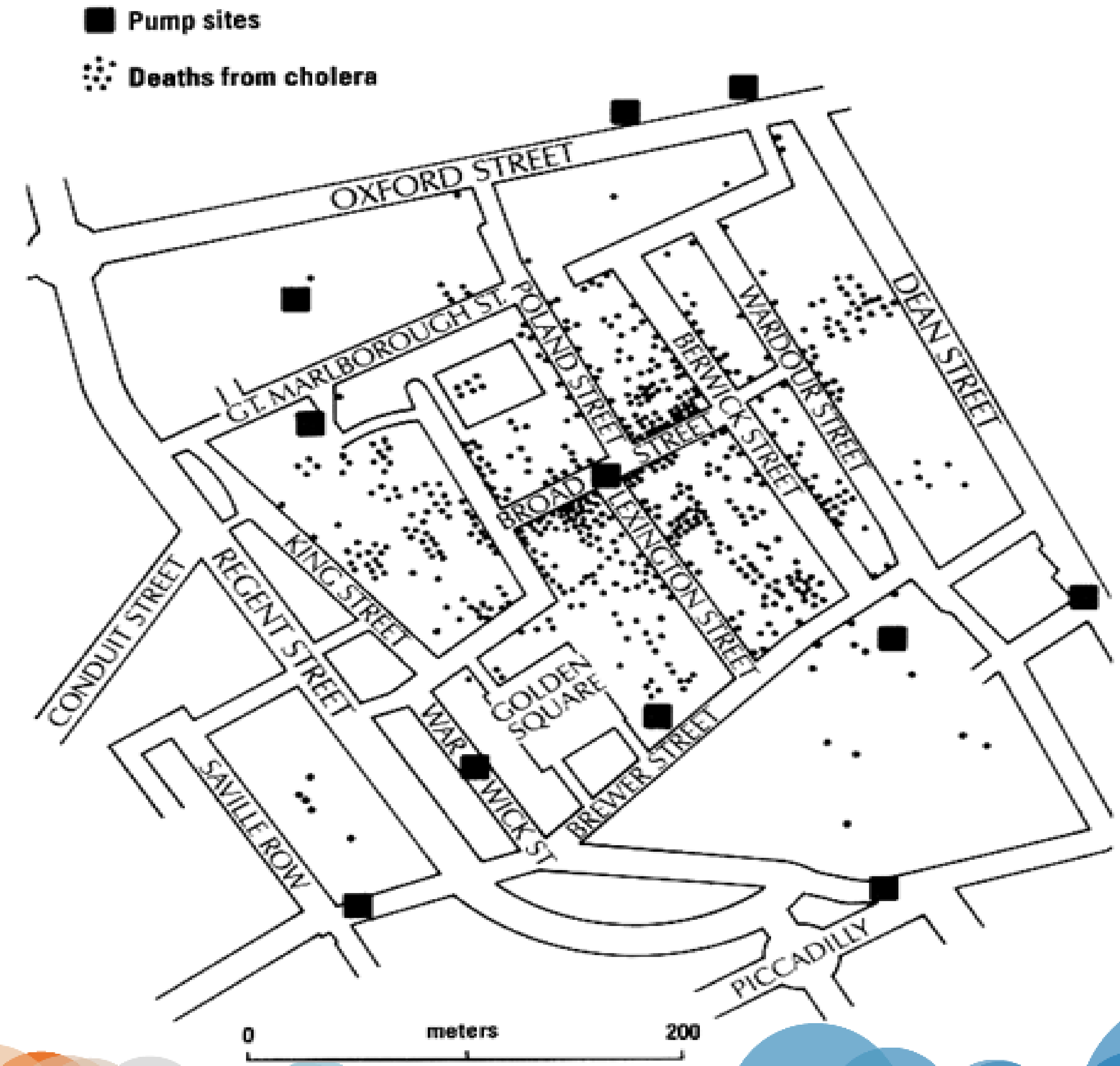
John Snow (1813-1858)
Cholera in London



Cholera Deaths



Cholera Deaths & H_2O Pumps



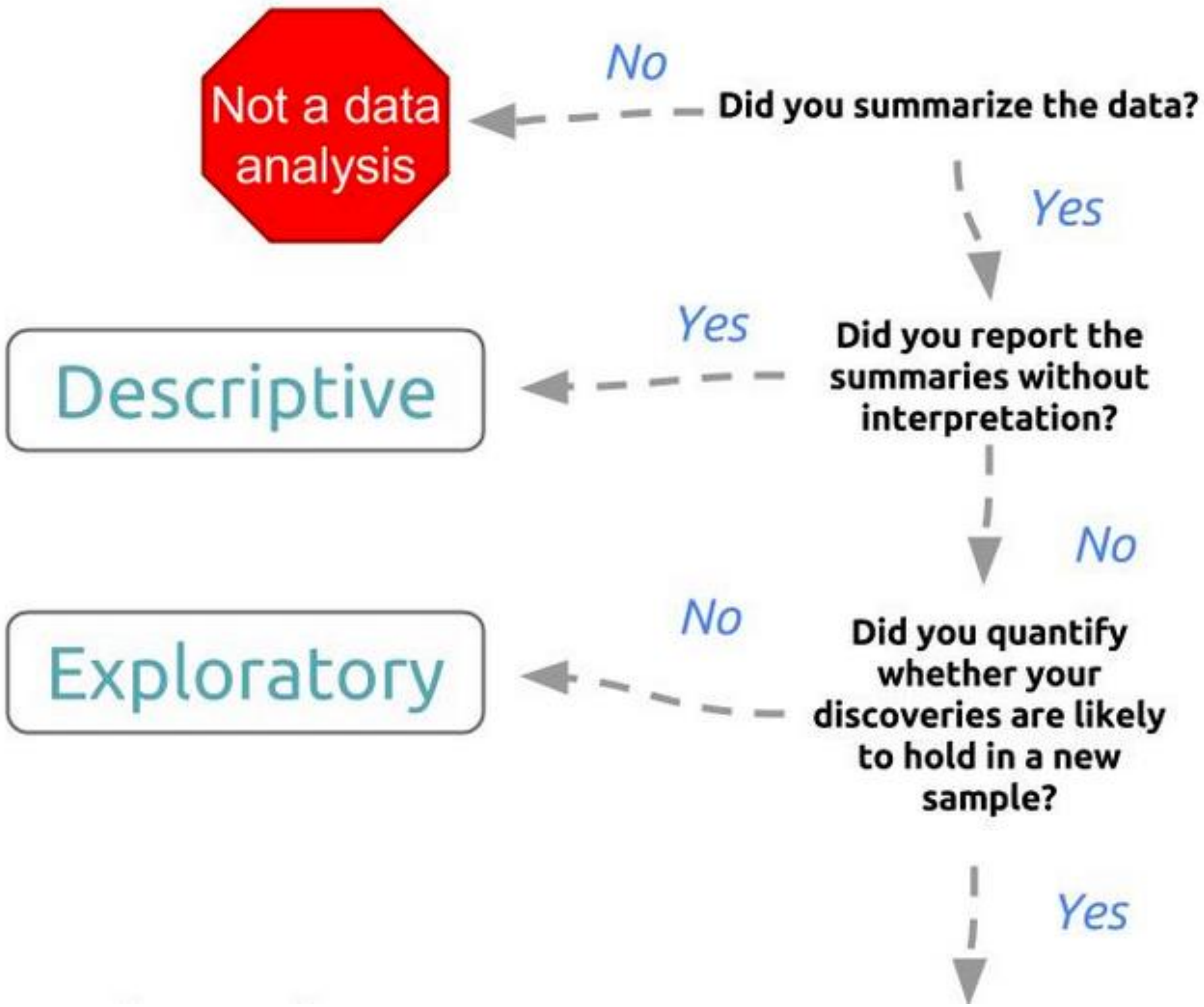
Mental Model of data analysis

1. The raw data come in
2. Format suitable for analysis
3. Then you conduct exploratory data analysis (EDA)
4. Once EDA is complete you can go to modeling the data (if necessary)
5. Once modeling and EDA is done, we can produce results
6. The results are then communicated

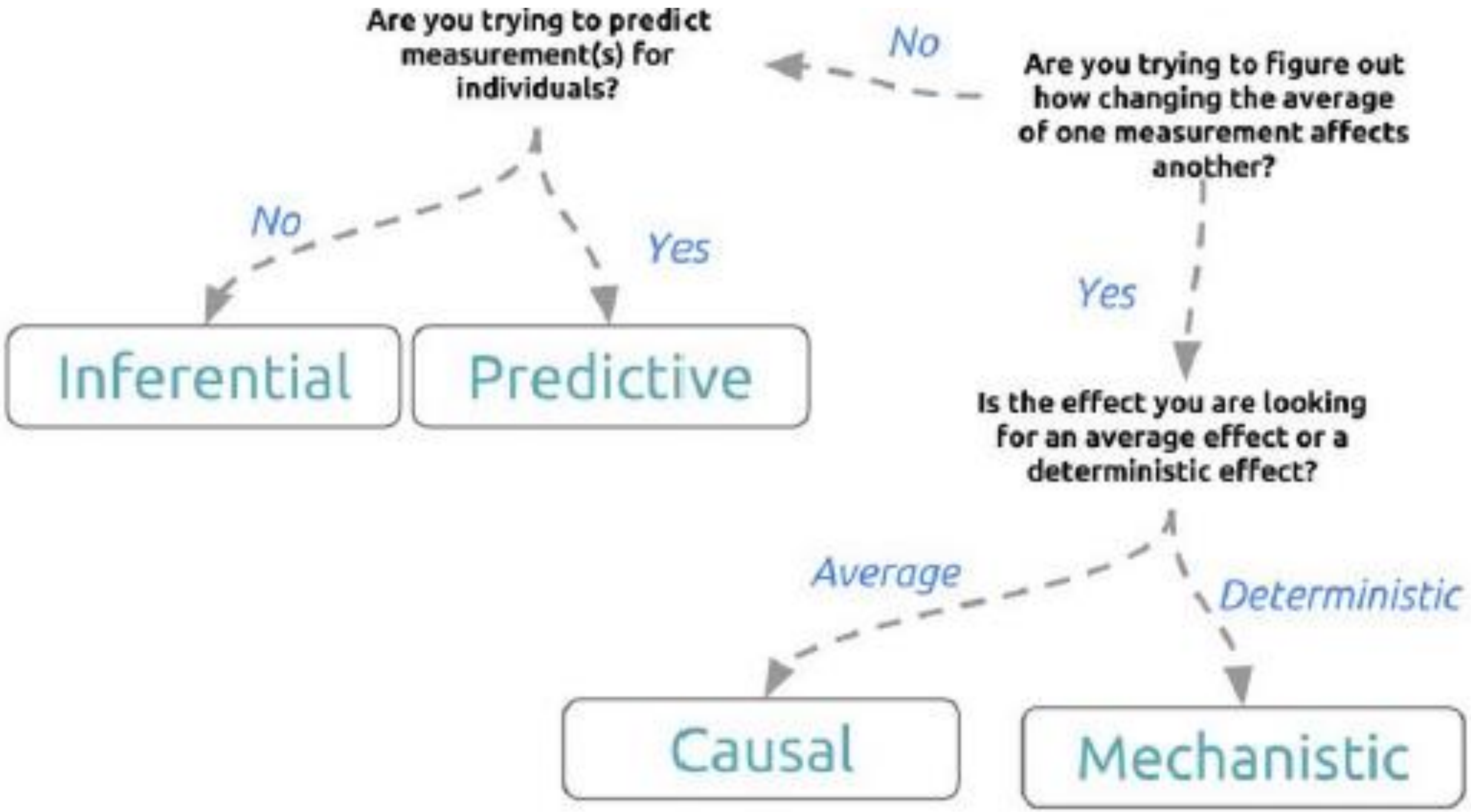
Mental Model Missing Key Aspects

- Context
- Resources
- Audience

The Data Analytic Question



The Data Analytic Question



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